CRICTRADE

The PSL Player & Team Trading Platform

Built on WireFluid Blockchain

Halal Trading Model	PSL 6 Teams 6 Tokens	Real-time Match Trading	AI-Powered Analytics

WireFluid Hackathon 2026 | March 24-26

Whitepaper v1.0 | Prepared by BlockVigil Engineering

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1. Executive Summary

CricTrade is a decentralized PSL (Pakistan Super League) team token trading platform built on the WireFluid blockchain. It transforms cricket fan engagement into a real-time financial experience where fans buy and sell team tokens whose economics are directly driven by on-field performance. Unlike traditional betting platforms, CricTrade operates on Islamic trading principles (Tijarat) — users own real digital assets, can exit at any time, and even losing positions retain value through performance rewards and upset bonuses.

The platform introduces three groundbreaking mechanisms: **Dynamic Performance-Linked Sell Tax** (underperforming teams become expensive to abandon, creating natural price stability), **Upset Vault Rewards** (massive bonus payouts when underdogs defeat favorites, rewarding conviction), and **AI-Powered Cricket Analytics** (providing traders with institutional-grade analysis of matchups, form, and value opportunities).

The platform earns revenue through trading fees on every transaction — regardless of match outcomes. This mirrors the profitable economics of betting platforms while maintaining full Sharia compliance. All treasury and reward mechanisms are governed by immutable smart contracts on WireFluid, ensuring transparency and zero platform risk.

$0.01	5 sec	6	10M PKR
WireFluid Tx Fee	Transaction Finality	PSL Team Tokens	Hackathon Prize Pool

2. Problem Statement

2.1 Why Betting is Haram in Islam

Traditional sports betting is classified as **Qimar (gambling)** in Islamic jurisprudence and is explicitly prohibited in the Quran (Surah Al-Maidah 5:90-91). The fundamental characteristics that make betting haram are:

Characteristic	Why It Is Haram
No Asset Ownership	Bettor receives nothing tangible — just a promise of payout based on outco
Binary Loss	If the predicted outcome is wrong, 100% of the stake is lost with nothing ren
Chance-Based	Outcome depends on uncontrollable external events, not productive effort
Zero-Sum	One party must lose for another to gain — no collective value is created
No Exit Option	Once a bet is placed, the participant cannot exit or modify their position

2.2 The Market Gap

Cricket fans in Pakistan (and globally across 2.5 billion cricket followers) have enormous passion for the sport but extremely limited avenues for financial engagement. The existing options are either:

1. Illegal betting apps — Widely used in Pakistan despite being both illegal under Pakistani law and haram in Islam. Estimated $50B+ annual market in South Asian cricket betting, mostly unregulated.

2. Fantasy cricket (Dream11 etc.) — Technically not available in Pakistan. Even where available, Islamic scholars debate its permissibility since the entry fee + prize pool structure mirrors gambling.

3. Failed crypto attempts — Rario ($120M raised, shut down), FanCraze ($117M raised, faded), Cricket Foundation (token went to zero). All failed because they sold speculative static collectibles, not real-time engagement tools.

The gap: There is no Sharia-compliant, legally defensible, real-time financial engagement platform for cricket fans. CricTrade fills this gap by converting the betting model into a trading model — preserving the excitement and economic incentives while replacing every haram element with a halal alternative.

3. The CricTrade Solution

3.1 Core Mechanics

At the start of each PSL season, six team tokens are created — one for each franchise. These tokens trade on bonding curves deployed on WireFluid. The price of each token is determined purely by market supply and demand, but the platform's economic mechanisms ensure that real-world cricket performance is the PRIMARY driver of demand, not speculation.

Step	What Happens
1. Season Start	Six team tokens launch on bonding curves at equal base prices (Fair Launch Phase)
2. Pre-Match	Fans analyze matchups, buy team tokens based on their cricket knowledge and AI insights
3. Live Match	Real-time trading continues. Token prices move with match momentum as fans react to events
4. Match End	Performance scores calculated. Sell taxes recalibrate. Reward pools distribute to holders
5. Upsets	When underdogs win, Upset Vault releases massive bonus rewards to underdog token holders
6. Season End	Grand Prize distributed. Tokens become season collectibles. New season launches fresh tokens

3.2 Per-Match Token Lifecycle

Every PSL match creates a fresh trading cycle. Before the match, fans take positions based on analysis. During the match, prices fluctuate as wickets fall, runs are scored, and momentum shifts. Each swing generates trading volume, and every trade generates platform fees. After the match, the performance score oracle updates all team metrics, recalibrating the dynamic sell tax and distributing performance rewards.

3.3 Asymmetric Bonding Curve

CricTrade uses an asymmetric bonding curve — the buy curve is steeper than the sell curve. This means buying pushes the price up faster, but selling brings it down more slowly. The spread between the two curves is the platform's structural protection against manipulation. Quick flips result in guaranteed losses due to the asymmetry, while longer holds allow the curves to converge, reducing the spread penalty.

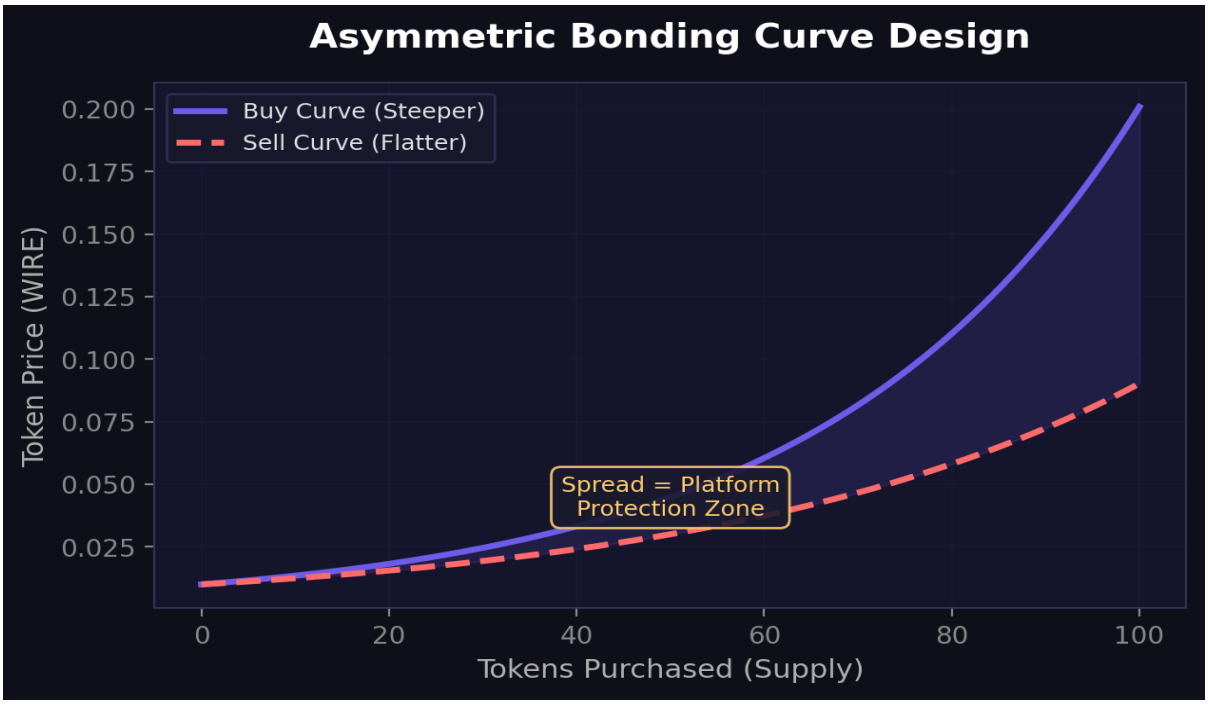

Fig 1: Asymmetric bonding curve — buy curve is steeper than sell curve, creating a protection spread

4. Economic Model

4.1 Dynamic Performance-Linked Sell Tax

The most innovative feature of CricTrade is the dynamic sell tax that adjusts based on real team performance. The best-performing team in the PSL has the lowest sell tax (2%), making it easy and cheap to trade. The worst performing team has the highest sell tax (15%), making it expensive to abandon. This creates a natural economic incentive to HOLD underperforming teams rather than panic-sell, stabilizing prices and creating real trading decisions based on analysis rather than impulse.

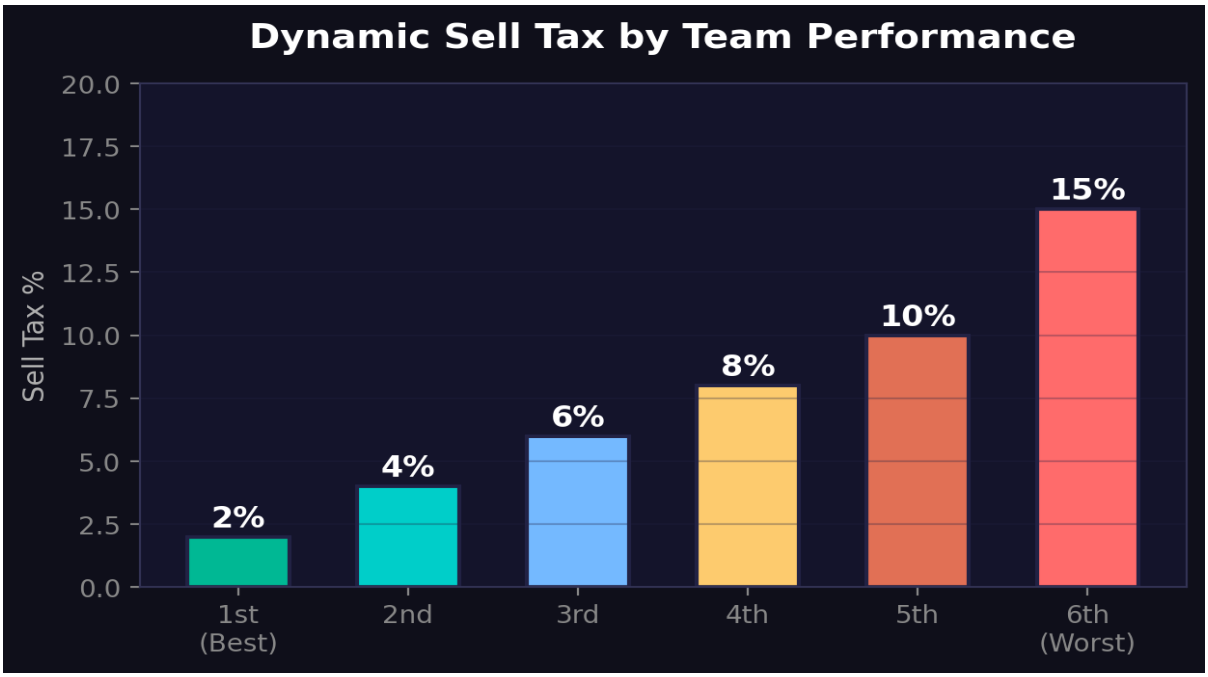

Fig 2: Dynamic sell tax scales inversely with team performance — worst team has 15% exit cost

The sell tax recalculates automatically after every match based on the team's updated Performance Score:

Metric	Weight	Description
Points Table Position	40%	Direct league standing (1st = highest score, 6th = lowest)
Net Run Rate (NRR)	20%	Overall tournament NRR, normalized to 0-100 scale
Last 3 Match Form	20%	Win = 20 points each, Loss = 0 points. Maximum 60 points.
Player Availability	20%	Key player injuries, bans, and replacement quality index

4.2 Performance Reward Distribution

After each match, the Performance Reward Pool (25% of all collected fees) distributes to token holders based on their team's league position. Top-ranked team holders receive the largest share, creating a passive income stream for those who correctly identified strong teams early. Even bottom-ranked team holders receive a small allocation, ensuring no participant is completely shut out.

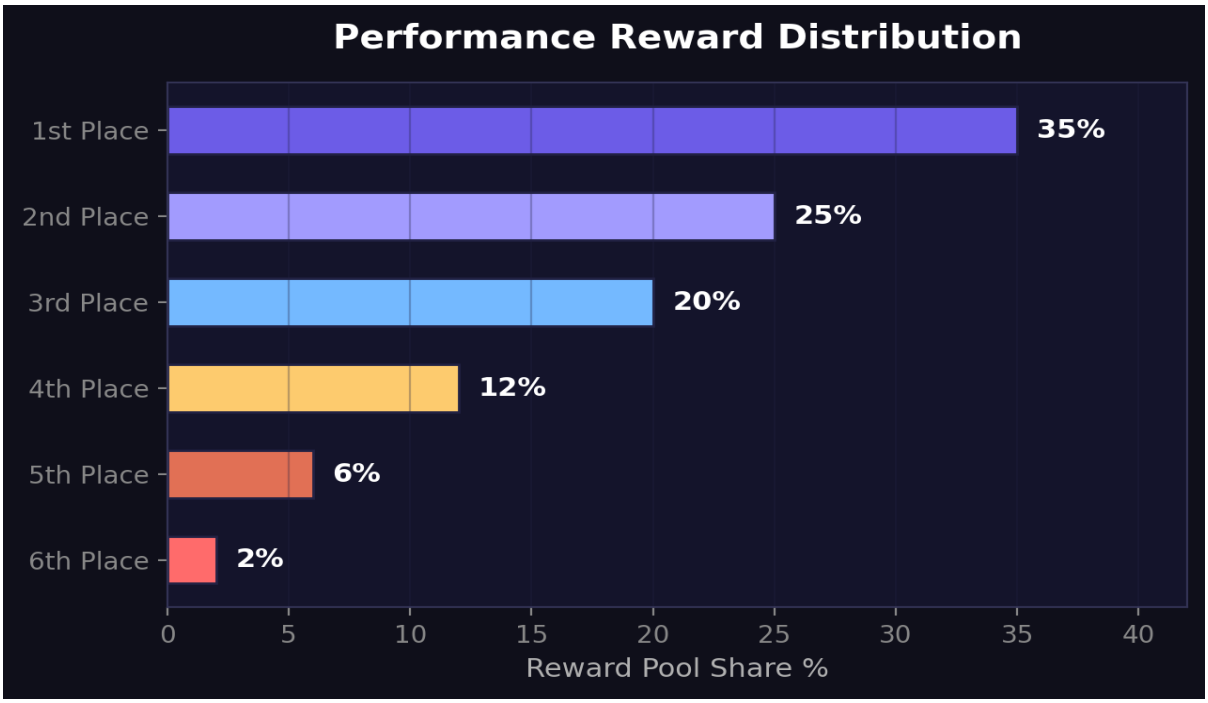

Fig 3: Performance reward distribution — 1st place holders receive 35% of the reward pool

4.3 Fee Revenue Split

Every trade on CricTrade generates a fee. The buy fee is a flat 2%, while the sell tax ranges from 2% to 15% depending on the team's performance ranking. All collected fees are split across five pools, each serving a critical function in the platform's economy:

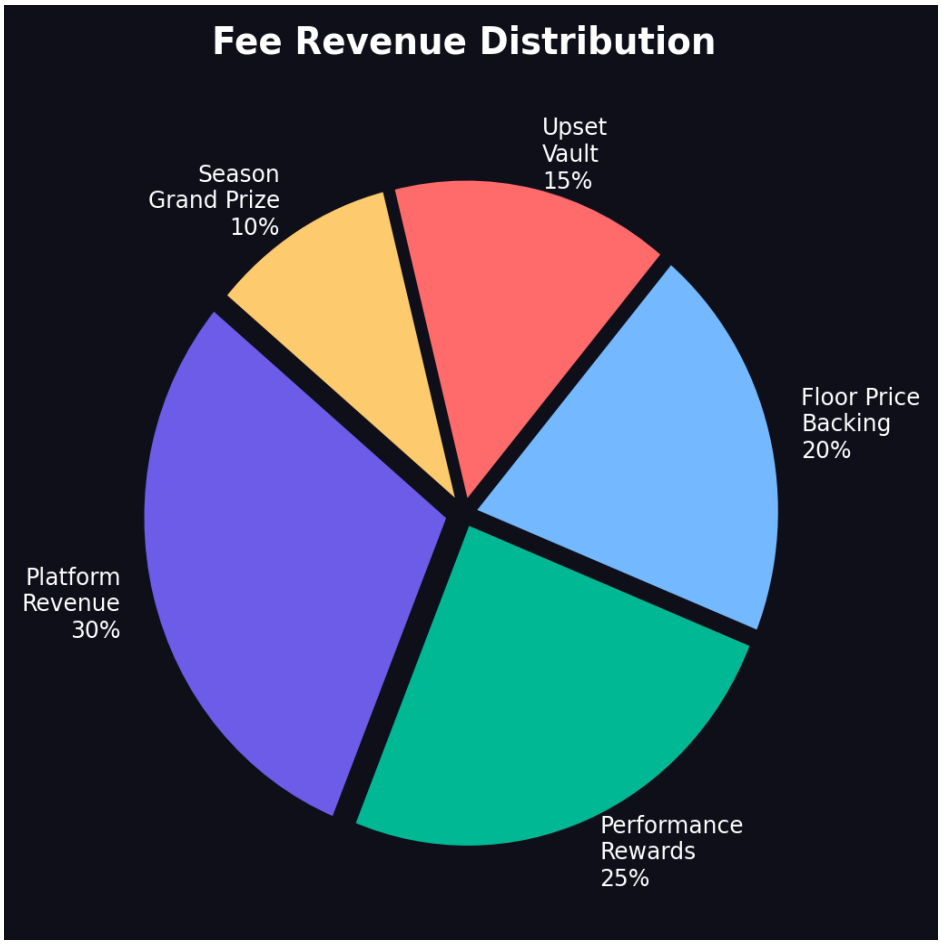

Fig 4: Revenue distribution — Platform retains 30%, remaining 70% powers the ecosystem

4.4 Complete Money Flow

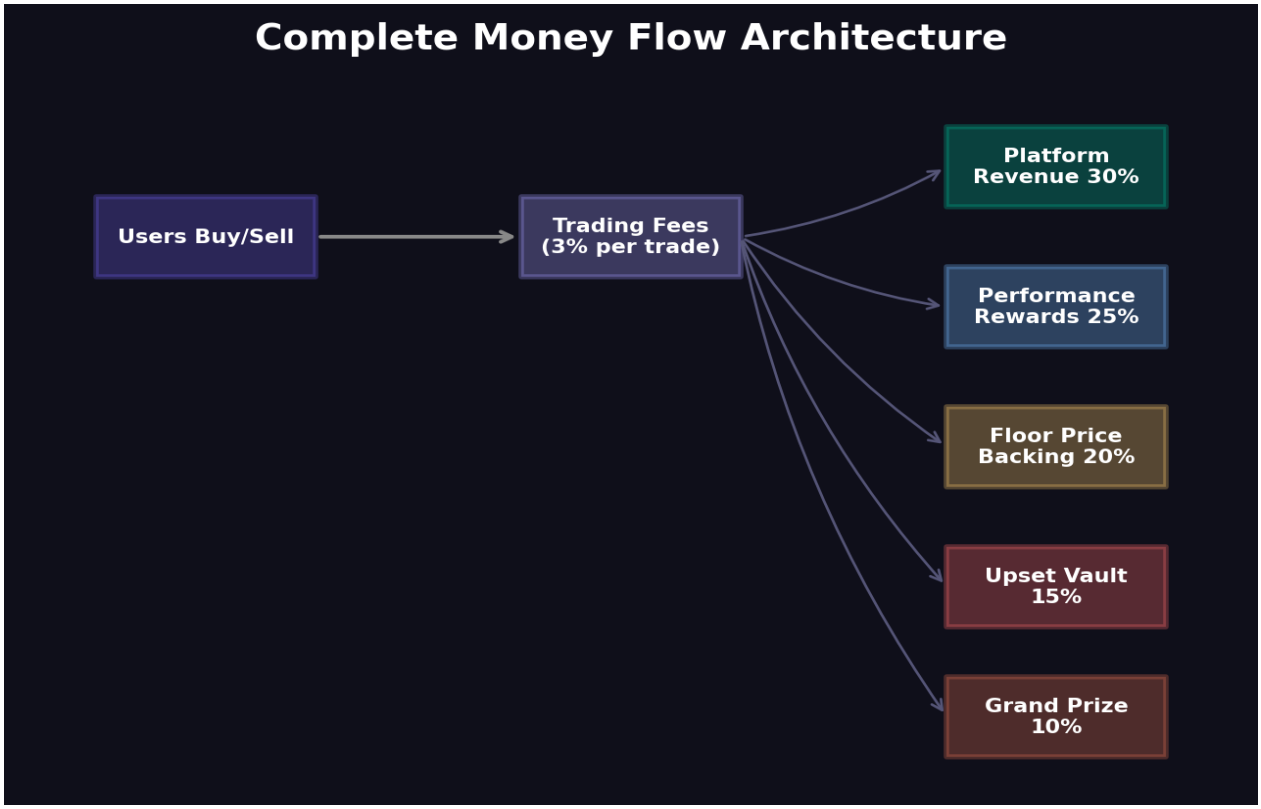

Fig 5: End-to-end money flow from user trades through fee distribution

The key insight: **the platform earns revenue on EVERY trade regardless of match outcome.** When teams win, their fans trade actively. When teams lose, sell taxes are higher, generating more fee revenue per transaction. More dramatic matches produce more price swings, more trades, and more fees. The platform's incentives are perfectly aligned with maximum fan engagement — not with any particular team winning.

5. Upset Vault System

The Upset Vault is CricTrade's signature innovation — a mechanism that creates explosive reward events when underdogs defeat favorites. It is designed to reward conviction and create viral moments that drive platform growth.

5.1 Upset Multiplier Mechanics

The Upset Score is calculated as the difference between the winning team's sell tax and the losing team's sell tax. When a team with a high sell tax (poor performer) defeats a team with a low sell tax (strong performer), the gap represents the magnitude of the upset. This gap determines the reward multiplier:

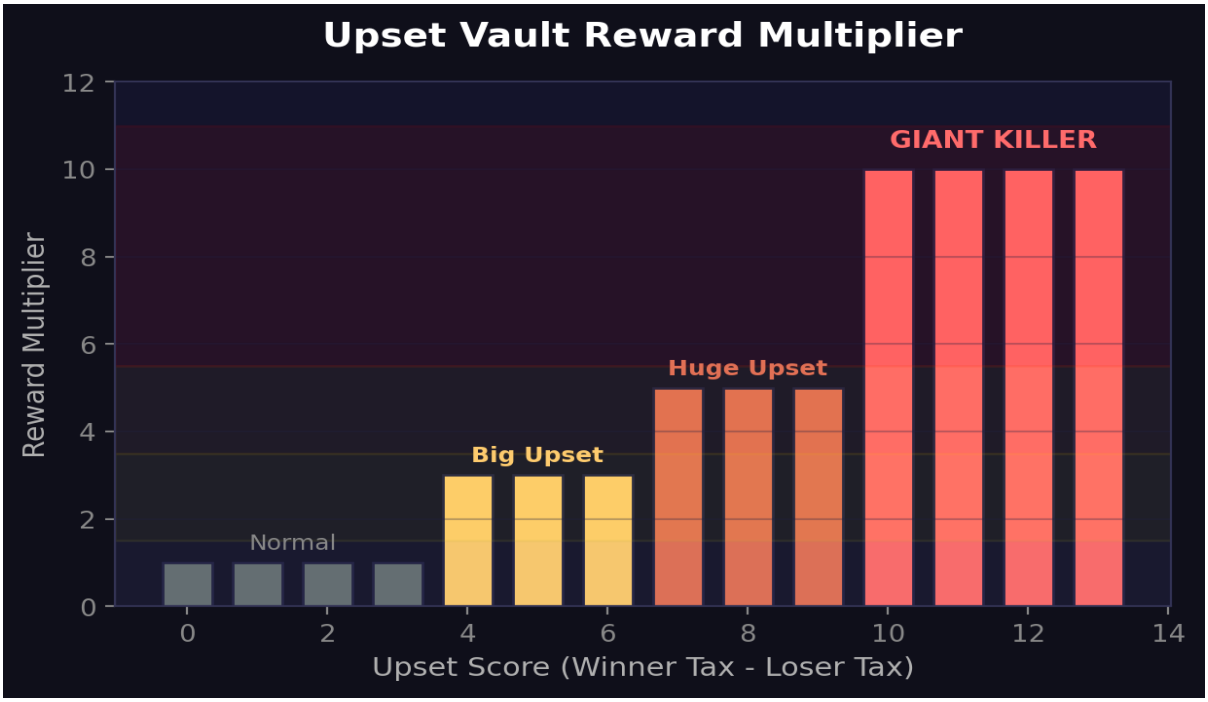

Fig 6: Upset score determines reward multiplier — Giant Killer events (10x) are the most rewarding

5.2 Vault Build & Release Cycle

The Upset Vault continuously accumulates 15% of all platform fees. When expected results occur (strong teams beating weak teams), the vault grows larger. This creates anticipation — the longer the vault builds without an upset, the larger the eventual payout will be. When an upset finally triggers, the vault releases a percentage proportional to the upset magnitude:

Upset Level	Multiplier	Vault Release %	Example Scenario
Normal (0-3)	1x	0% (regular rewards)	3rd beats 4th
Big Upset (4-6)	3x	15% of vault	5th beats 2nd
Huge Upset (7-9)	5x	30% of vault	6th beats 3rd

GIANT KILLER (10-13)	10x	60% of vault	6th beats 1st

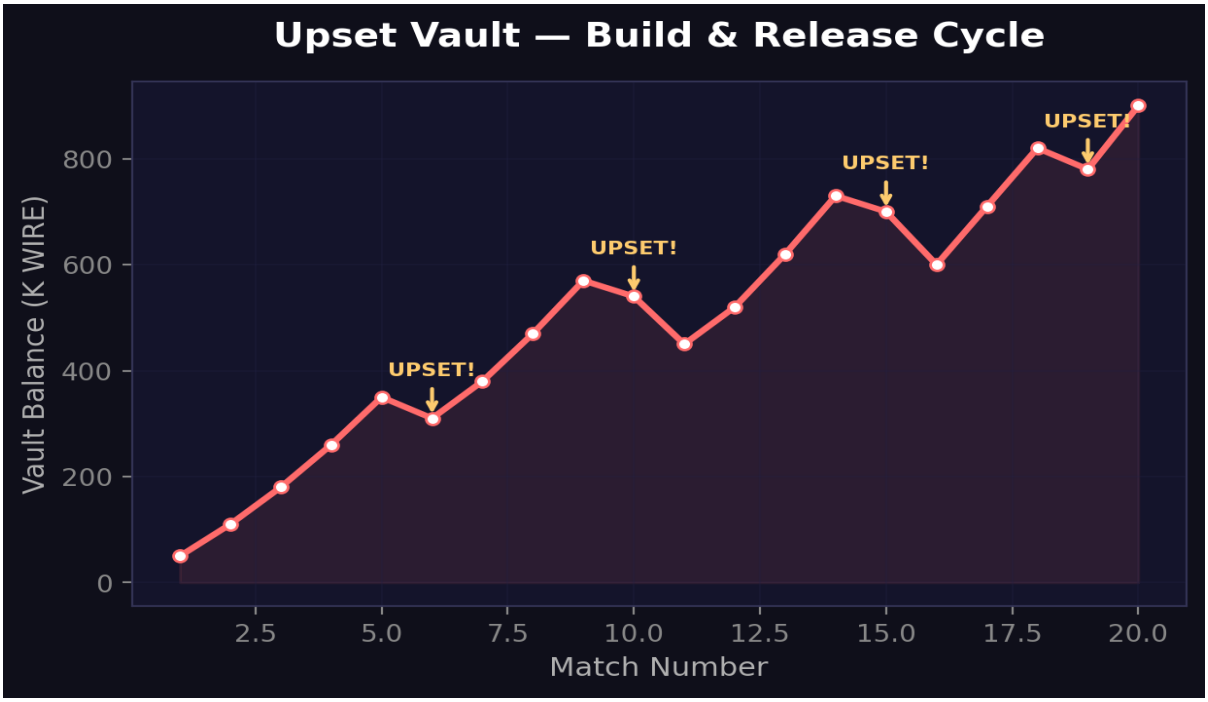

Fig 7: Upset Vault builds during expected results and releases on upsets — creating explosive reward moments

5.3 Giant Killer Scenario Walkthrough

Consider this real-world scenario: It is Match Day 15 of PSL 2026. Islamabad United sits at the top of the table with a 2% sell tax. Quetta Gladiators are bottom with 15% sell tax. The Upset Vault has accumulated 800,000 WIRE over 14 matches of mostly expected results.

A small group of astute fans noticed that Quetta's bowling attack matches up well against Islamabad's batting lineup. They quietly buy $QG tokens at 0.02 WIRE — near the floor price. Most of the market ignores Quetta.

Result: Quetta wins. Upset Score = 15 - 2 = 13 (GIANT KILLER). The Upset Vault releases 60% of its balance = 480,000 WIRE, plus the normal match reward of 50,000 WIRE with a 10x multiplier. The 200 remaining Quetta holders receive approximately 2,650 WIRE each. Those who bought at 0.02 WIRE spent 100 WIRE to buy tokens and received 2,650+ WIRE in upset rewards — a 26x return on a single match, not counting the token price appreciation.

This moment becomes viral. Screenshots of 26x returns spread across social media. New users flood the platform. Trading volume spikes. The vault begins refilling for the next upset cycle.

6. Anti-Whale Protection

CricTrade implements six layers of whale protection to prevent market manipulation and protect retail traders:

Layer	Mechanism	Effect on Whale
1. Asymmetric Curve	Sell curve flatter than buy curve	Quick flips guarantee losses due to spread
2. Progressive Sell Tax	12% within 5 min, 8% within 30 min, 5% within 2 hrs, 3% after	Fast dumps are extremely expensive
3. Max Transaction Limit	1% of total supply per sell + 60 second cooldown	100+ transactions needed for full exit
4. Performance Floor	Treasury-backed minimum price. Auto-buy at floor level	Cannot crash price below floor
5. Protection Pool	Whale sell taxes distributed to remaining holders	Whale dump = airdrop for holders
6. Circuit Breaker	15% drop in 5 min = 3 min pause. 25% in 15 min = 10 min pause	Dump momentum broken, panic prevented

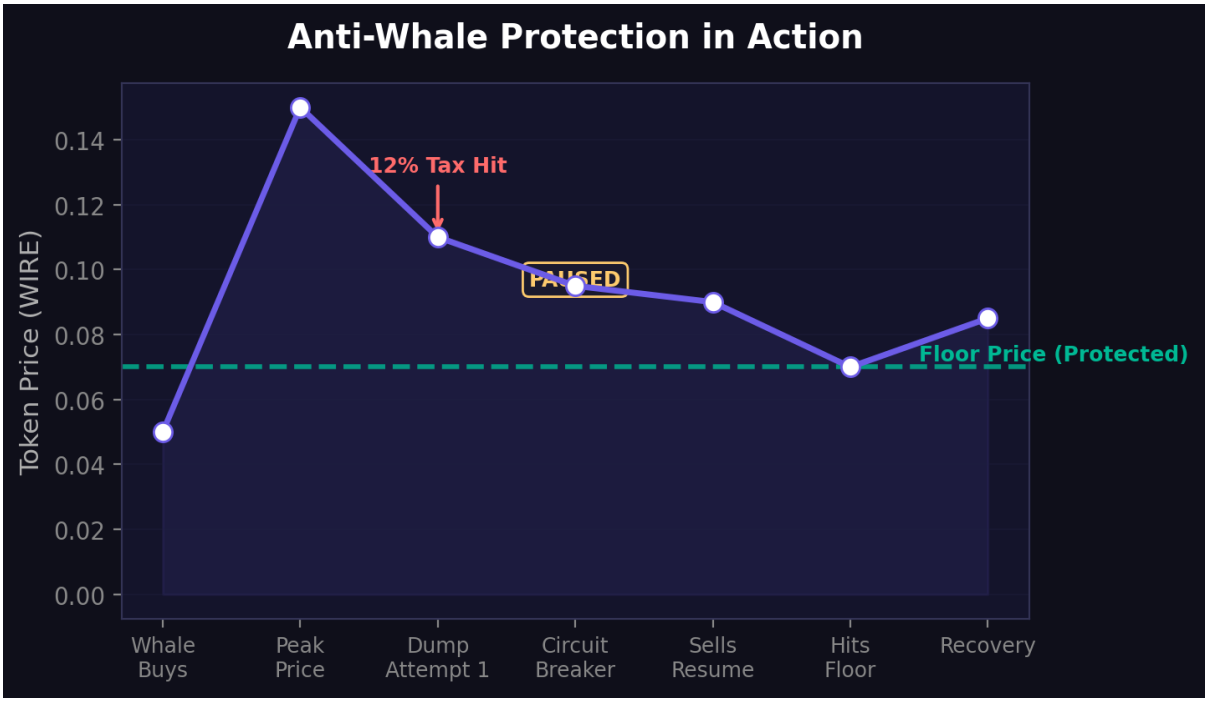

Fig 8: Whale dump scenario — price stabilizes at floor after circuit breaker and tax penalties

7. Security & Exploit Protection

CricTrade's security model addresses 12 identified attack vectors across smart contract, economic, and operational layers. Each exploit has been analyzed and countered with specific mechanisms:

Security & Exploit Protection Matrix

Attack Vector	Risk	Protection
Multi-Wallet Sybil	HIGH	Phone/CNIC + AI
Transfer Tax Dodge	CRITICAL	Transfer = Sell
Flash Loan	HIGH	24hr Delay + Hold Req
Whale Dump	HIGH	Circuit Breaker + Floor
Spread All Underdogs	MEDIUM	Diminishing Returns
Front-Running	HIGH	Commit-Reveal Batch
Oracle Manipulation	CRITICAL	Multi-Source + Multisig
Last-Min Reward Grab	MEDIUM	Match-Start Snapshot

Fig 9: Complete security exploit matrix — all identified attack vectors with corresponding protections

Key Security Mechanisms:

Transfer = Sell: All token transfers incur the same dynamic sell tax as market sells. This eliminates the most dangerous exploit — transferring tokens to a secondary wallet to avoid sell tax. Receiving wallets inherit zero hold duration.

Flash Loan Defense: Rewards distribute in a 24-hour claim window, not within the same block. Combined with hold duration requirements, flash loan attacks become mathematically unprofitable.

Commit-Reveal Trading: During live matches, orders are submitted in encrypted form (commit phase) and revealed simultaneously in 30-second batches. This eliminates front-running and MEV extraction.

Multi-Source Oracle: Match data is verified by 2-out-of-3 independent sources (CricAPI, Cricbuzz, ESPN). If sources conflict, trading pauses for manual review via multisig.

Match-Start Snapshot: Reward eligibility is determined at the time of the first ball, not match end. This prevents last-minute reward-grabbing.

8. AI Integration

AI is not an add-on feature in CricTrade — it is a core component that makes the platform's complex economic dynamics navigable for users. Without AI-powered analysis, the dynamic sell tax, upset vault, and multi-team portfolio management would be overwhelming for casual fans. The AI serves as each user's personal cricket trading analyst.

8.1 Pre-Match Analysis

Before each match, the AI generates comprehensive research reports using a RAG (Retrieval-Augmented Generation) pipeline. Historical PSL ball-by-ball data (21,000+ matches from Cricsheet) is embedded in a vector database. The AI retrieves relevant historical matchups, venue statistics, and bowling-batting head-to-head records to produce institutional-grade analysis:

> *"Quetta's sell tax is at 15% — highest this season. However, their bowling attack has a combined average of 19.3 against left-handed batsmen, and Islamabad's top 4 includes 3 left-handers. In similar matchups across PSL 8-10, underdogs won 34% of the time. The Upset Vault is at 800K WIRE — the largest it has been this season. Risk/reward assessment: High risk, extraordinary potential reward. Suggested position: Small allocation (5% of portfolio) with 24+ hour hold for full upset reward eligibility."*

8.2 Live Match Intelligence

During matches, the AI processes ball-by-ball data and provides real-time trading signals with reasoning. It monitors momentum shifts, identifies overreactions in token prices, and alerts users to opportunities. The AI also tracks the Upset Score trajectory — estimating how likely an upset payout is as the match progresses.

8.3 Portfolio Management

Post-match, the AI generates portfolio performance reports, analyzes entry/exit timing quality, and recommends rebalancing strategies. It factors in remaining fixtures, strength of schedule, and sell tax trajectories to advise whether to hold, reduce, or increase positions in each team.

9. WireFluid Network Advantages

CricTrade specifically requires WireFluid's capabilities. The same product would not work on Ethereum, Polygon, or Solana due to the unique combination of requirements:

WireFluid Feature	Why CricTrade Needs It	Alternative Chain Limitation
$0.01 Transaction Fees	Fans make 10-50 micro-trades per match. $0.01 fees keep trading viable.	Ethereum: $5-50 per tx would destroy the micro-trade model entirely.
5-Second Finality	Token price must update within one over (4 minutes). Instant finality ensures trades reflect live action.	Bitcoin: 10 min blocks. Even Polygon's 2 sec blocks have reorg risk.
EVM Compatibility	Standard Solidity smart contracts, Hardhat tooling, MetaMask integration. No new language to learn.	Solana: Requires Rust/Anchor. Cosmos native: Requires Go modules.
CometBFT Consensus	No mempool means no MEV front-running. Validators order transactions fairly.	Ethereum: MEV bots extract value from every prediction/trading platform.
IBC Cross-Chain	Future expansion to users on other Cosmos chains. Asset bridging.	EVM chains: Rely on fragile bridges with security risks.

10. Halal Compliance Framework

CricTrade is designed from the ground up to comply with Islamic finance principles. Every mechanism was evaluated against the criteria that distinguish halal trade (Tijarat) from haram gambling (Qimar):

CricTrade vs Traditional Betting

Feature	Traditional Betting	CricTrade
Asset Ownership	None	Full Token Ownership
Loss Scenario	100% Loss	Partial + Rewards
Exit Anytime	No	Yes
Skill Based	Mostly Luck	Analysis + Knowledge
Sharia Compliance	Haram (Qimar)	Halal (Tijarat)
Platform Risk	Against Users	Zero Directional

Fig 10: Side-by-side comparison — CricTrade satisfies every Islamic trading requirement

Detailed Halal Justification:

Islamic Principle	CricTrade Implementation
Ownership (Milkiyat)	User receives ERC-20 tokens that are recorded on the blockchain. They hold full ownership of a real digita
Voluntary Exit (Ikhtiyar)	Users can sell their tokens at any time. There is no lock-in period. The sell tax is a cost of trading, not a re
No Total Loss	Token value may decrease, but floor price mechanism ensures non-zero value. Reward distributions provi
Knowledge-Based (Ilm)	Successful trading requires cricket knowledge, analytical ability, and market understanding — productive s
No Gharar (Excessive Uncertainty)	All mechanics are transparent and defined in smart contracts. No hidden terms, no counterparty risk, no u
Collective Value	Unlike betting where losers fund winners, CricTrade creates a shared engagement economy. Platform fee

11. Live Match Trading Flow

The most exciting aspect of CricTrade is the live match trading experience. As the match unfolds over by over, token prices fluctuate in real-time based on fan reactions, AI signals, and match events. The following charts illustrate a typical live match trading session:

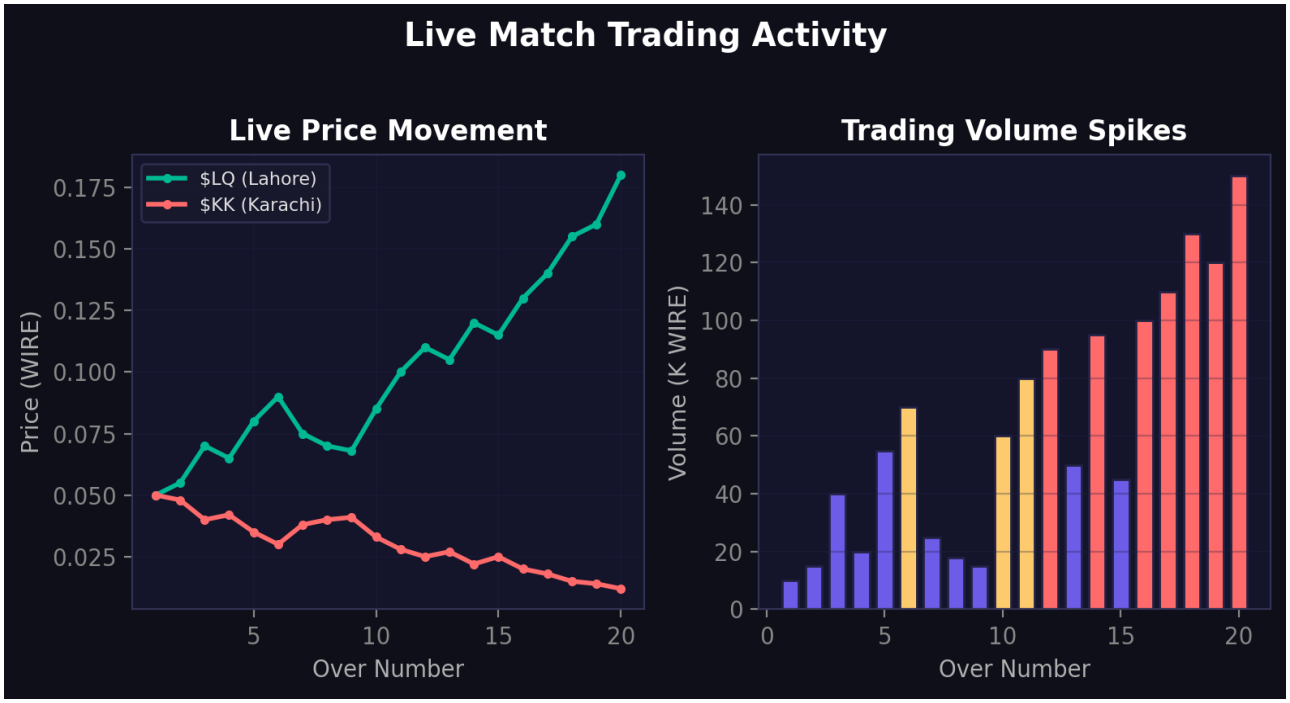

Fig 11: Live match price movement and volume — token prices react to match events; volume spikes on key moments

Match Trading Timeline:

Phase	Time	Activity
Pre-Match Window	2 hours before	Fans analyze teams, AI reports published, initial positions taken
Toss & Team Announce	30 min before	Sell pause until teams announced. Then trading resumes with full info
Powerplay (Overs 1-6)	Match start	High volatility. Boundaries and wickets cause rapid price swings
Middle Overs (7-15)	Mid-match	Price trends establish. Momentum becomes clearer. Smart repositioning
Death Overs (16-20)	Late match	Maximum trading volume. Outcome becoming clear. Massive price movement
Match Settlement	1 min after end	Oracle updates scores. Sell taxes recalibrate. Rewards queue for 24hr claim
Post-Match Trading	24 hours after	Reflection trades. Position building for next match. Reward claims open

12. Season Lifecycle

Phase	Duration	Details
Token Launch	Day 1	Six team tokens mint via Fair Launch (24hr fixed-price phase, max 10K WIRE per wallet)
Bonding Curve Active	Day 2+	Price discovery begins. Sell taxes start at 5% for all teams
League Stage	~30 matches	Full trading cycle per match. Taxes/rewards recalibrate after each game
Playoff Qualification	After league	Eliminated team tokens: sell tax freezes. Qualified teams: tax drops to minimum 2%
Playoffs	4 matches	Trading intensity peaks. Upset Vault at maximum. Highest volume period
Final	1 match	Championship match trading. Maximum reward multipliers possible
Season Settlement	Post-final	Grand Prize Pool distributes. Champion holders get 50%. All tokens become collectibles
Off-Season	Until next PSL	Tokens remain tradeable as collectibles. Next season launches fresh token set

Season-End Grand Prize Distribution:

Placement	Prize Pool Share	Additional Reward
Champion Team Holders	50%	Champion NFT Badge (tradeable)
Runner-Up Holders	25%	Finalist NFT Badge
3rd/4th Place Holders	15%	Playoff NFT Badge
5th/6th Place Holders	10% (consolation)	Season Memorabilia Token

13. Technical Architecture (Summary)

Smart Contract Stack:

Contract	Purpose	Key Functions
`TeamTokenFactory.sol`	Creates team ERC-20 tokens with bonding curves	createTeamToken(), buy(), sell() calculateBuyPrice(), calculateSellPrice()
`PerformanceOracle.sol`	Receives and validates match performance data	updateMatchResult(), getPerformanceScore() getSellTaxRate(), verifyMultiSource()
`RewardDistributor.sol`	Manages all reward pools and distributions	distributeMatchRewards(), claimRewards() calculateUpsetBonus(), releaseVault()
`AntiWhale.sol`	Protection mechanisms	checkTransferLimits(), applyProgressiveTax() triggerCircuitBreaker(), getWalletTier()
`SeasonManager.sol`	Season lifecycle and grand prize management	startSeason(), endSeason() distributeGrandPrize(), fairLaunchPhase()

Technology Stack:

Layer	Technology
Blockchain	WireFluid (Cosmos SDK + EVM | CometBFT Consensus)
Smart Contracts	Solidity 0.8.x | Hardhat | OpenZeppelin
Backend	Node.js + Express | WebSocket for live data
Frontend	Next.js + React | wagmi/viem | TailwindCSS
AI Engine	Claude/GPT-4 API | RAG with Cricsheet data | Vector DB
Oracle Data	CricAPI (live) + Cricbuzz + ESPN (multi-source verification)
Storage	IPFS (AI report archival) | PostgreSQL (user data)

14. Conclusion & Vision

CricTrade represents a fundamentally new category of sports fan engagement — one that captures the economic excitement of betting platforms while operating entirely within Islamic trading principles. By combining performance-linked dynamic economics, upset reward mechanisms, anti-whale protections, and AI-powered analytics, the platform creates a self-sustaining ecosystem where every participant — fans, the platform, and the blockchain — benefits from increased engagement.

The timing is extraordinary. PSL Season 11 launches on March 26, 2026 — the final day of the WireFluid Hackathon. Pakistan has 27 million active crypto users, 80 million mobile wallet accounts, and an insatiable appetite for cricket. Yet no product exists at the intersection of halal finance, real-time cricket engagement, and blockchain. CricTrade fills this gap.

Future Roadmap:

Phase 1 (Hackathon): Core MVP — team tokens, bonding curves, basic dynamic sell tax, AI pre-match reports, WireFluid testnet deployment.

Phase 2 (Post-Hackathon): Full upset vault, circuit breaker, commit-reveal trading, multi-source oracle, mobile-responsive frontend, wallet verification system.

Phase 3 (Production): Mainnet launch, live PSL integration, partnership discussions with PSL franchises, IBC cross-chain expansion, Sharia compliance certification from recognized Islamic finance authority.

Phase 4 (Expansion): IPL, BBL, CPL, and international cricket coverage. Potential expansion to football (Premier League, FIFA World Cup). Multi-sport platform with unified trading economy.

CricTrade — Where Cricket Knowledge Becomes Financial Power.

Built on WireFluid | Powered by AI | Compliant with Islamic Finance

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